

Tutorial Exercises – Series 3

Procedures & Functions

Exercise 1

After each algorithm, display the values of a and b inside the procedure and in the main program.

Algorithm Test_ByValue Var a, b: Integer Procedure Modify(x: Integer; y: Integer) Begin x $\leftarrow$ x + 10 y $\leftarrow$ y + 20 Write("Inside procedure: x = ", x, ", y = ", y) EndProcedure Begin a $\leftarrow$ 5 b $\leftarrow$ 15 Call Modify(a, b) a $\leftarrow$ a + 1 b $\leftarrow$ b + 1 Write("In main program: a = ", a, ", b = ", b) End	Algorithm Test_ByAddress Var a, b: Integer Procedure Modify(Var x: Integer; Var y: Integer) Begin x $\leftarrow$ x + 10 y $\leftarrow$ y + 20 Write("Inside procedure: x = ", x, ", y = ", y) EndProcedure Begin a $\leftarrow$ 5 b $\leftarrow$ 15 Call Modify(a, b) a $\leftarrow$ a + 1 b $\leftarrow$ b + 1 Write("In main program: a = ", a, ", b = ", b) End
Algorithm Test_Mixed Var a, b: Integer Procedure Modify(x: Integer; Var y: Integer) Begin x $\leftarrow$ x + 10	Algorithm Test_all_ByAddress Var a, b: Integer Procedure Modify(Var x: Integer; Var y: Integer) Begin

<pre> y ← y + 20 Write("Inside procedure: x = ", x, ", y = ", y) EndProcedure Begin a ← 5 b ← 15 Call Modify(a, b) a ← a + 1 b ← b + 1 Write("In main program: a = ", a, ", b = ", b) End </pre>	<pre> x ← x + 5 y ← y * 2 Write("Inside procedure: x = ", x, ", y = ", y) EndProcedure Begin a ← 4 b ← 6 Call Modify(a, b) a ← a + 10 b ← b + 3 Write("In main program: a = ", a, ", b = ", b) End </pre>
---	--

Exercise 2

Create a procedure **Swap(a, b)** that exchanges the values of a and b.

- 1- Test it using pass by value : Do the values change in the main program?
- 2- Test it using pass by address (reference) : What happens now?

Exercise 3

- 1- Write a function **Increment(n)** that returns $n + 1$.
- 2- Write a procedure **IncrementRef(Var n)** that adds 1 directly to the variable.
- 3- Compare both behaviors when called with the same variable.

Exercise 4

- 1- Write a procedure **Compute(var a, var b, var sum, var avg)** that reads two integers and calculates their sum and average using pass by reference.
- 2- Display the results in the main program.

Exercise 5

- 1- Write a function **Square(x)** that returns the square of a number.
- 2- Write a program that reads an integer, calls the function, and displays the result.

Exercise 6

- 1- Write a function **Max(a, b)** that returns the larger of two integers.
- 2- Use it in a main program to compare two numbers.

Exercise 7

- 1- Write a function **IsEven(n)** that returns true if the number is even, otherwise false.
- 2- Use it in the main program to display whether a number is even or odd.

Exercise 8

- 1- Write a function **Factorial(n)** that returns $n!$.
- 2- Test it with several values and display the result.

Exercise 9

Write two functions:

- 1- **CelsiusToFahrenheit(c)** : returns the equivalent in Fahrenheit ($F \leftarrow (9 / 5) * C + 32$)
- 2- **FahrenheitToCelsius(f)** : returns the equivalent in Celsius ($C \leftarrow (5 / 9) * (F - 32)$)
- 3- Use both in a main program.

Exercise 10

- 1- Write a function **SumN(n)** that returns the sum of numbers from 1 to n.
- 2- Display the result in the main program.

Exercise 11

- 1- Write a function **Square(x)** and another function **Cube(x)**.
- 2- Write a program that reads a number and displays both its **square** and **cube** using those functions.

Exercise 12

Write an algorithm that reads two integers a and b, calculates their **sum** and **average**, and displays both results.

- 1- Write a **function** Sum(x, y) that returns the sum of the two numbers.
- 2- Write a **function** Average(x, y) that calls the Sum function to compute the average.
- 3- In the main program, call the Average function and display both the sum and the average.

Exercise 13

- 1- Write an algorithm that reads three grades of a student.
- 3- A function **ComputeAverage** computes the average of the three grades.
- 4- A function **IsPassed** calls **ComputeAverage** and returns **TRUE** if the student passes (average ≥ 10), or **FALSE** otherwise.
- 5- The main program displays the result accordingly.